

Robert Lacheta

Full-Stack Engineer | React, Node.js & Geospatial Data Tooling

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PROFESSIONAL SUMMARY

Software engineer with 4+ years of commercial experience building and maintaining TypeScript web applications, plus hands-on full-stack and geospatial backend experience. Proficient in React, JavaScript, TypeScript, client-side state management, asynchronous workflows, micro-frontends, REST APIs, testing and CI/CD. Built Map Data Quality Lab, a source-aware platform for preparing, validating, reviewing and serving OSM-derived GeoJSON data through Node.js/Express services and a Python geospatial worker. Experienced in international agile teams; English B2 professional working proficiency.

TECHNICAL SKILLS

Frontend: JavaScript, TypeScript, React, Redux, Zustand, RxJS, Vite, HTML5, CSS3, SPA, UI/UX principles, DOM APIs, Shadow DOM
Backend & APIs: Node.js, Express.js, REST, Python, FastAPI, SQL (basic), request validation, asynchronous tasks, API contracts
Geospatial Data: GeoJSON, OpenStreetMap (OSM), Overpass, MapLibre GL JS, PMTiles, MVT/vector tiles, OSMnx, GeoPandas, Shapely, QGIS, WMS/WFS
Architecture: Micro-frontends, state management, side effects, software design patterns, frontend/backend integration, caching, data provenance
Testing & Delivery: Vitest, Jest, React Testing Library (RTL), Git, GitHub Actions, CI/CD, Docker, Jenkins, Jira, code review
Additional: Google Cloud fundamentals, C++ fundamentals, C# basics, English B2

PROFESSIONAL EXPERIENCE

Frontend Developer | ALAN Systems

Apr 2022 - Present

Large-scale betting platform and customer-retention systems | International agile team

- Develop and maintain complex customer-facing modules with TypeScript, React and Redux.
- Contribute to a micro-frontend architecture, including project configuration, Shadow DOM interoperability and integration with backend microservices.
- Work across analysis, design, implementation, testing, release and maintenance as business and product requirements evolve.
- Manage asynchronous application state, side effects and DOM interactions, and optimize frontend performance for responsive workflows.
- Write unit and component tests with Vitest/Jest and React Testing Library (RTL); participate in code reviews and CI/CD-based releases.
- Implemented GrowthBook for A/B testing and feature-flag management.

Sales Representative | Epax

Feb 2020 - Mar 2022

Advised customers on computers, servers and office hardware; built long-term client relationships.

Consumer Banker | Alior Bank

Nov 2018 - Jan 2020

Managed customer relationships and coordinated service across multiple branches.

Customer Consultant / Digital Marketing & Sales Manager | Bike System

Mar 2012 - Oct 2018

Managed PrestaShop operations, customer service, digital campaigns, SEO and Google Analytics.

SELECTED ENGINEERING PROJECTS

Map Data Quality Lab | Personal Geospatial Data Tooling Project

Jul 2026 - Present

React, TypeScript, Node.js, Express.js, Python, GeoJSON, OSM, MapLibre GL JS, PMTiles, Docker

- Designed and built a source-aware geospatial data provider that prepares OSM/Overpass infrastructure layers for parameterized areas of interest and normalizes them into versioned GeoJSON contracts across nine core domains.
- Implemented a hybrid Node.js/Express/TypeScript REST service and Python worker using OSMnx, GeoPandas and Shapely, with deterministic caching, request validation, source provenance, confidence and readiness metrics.
- Built a React/TypeScript MapLibre inspection UI with PMTiles/MVT range reads, layer catalog, feature details, source limitations and a persistent data-quality issue review workflow.
- Created reproducible offline verification and CI/CD gates with Python and TypeScript tests; the committed Rybnik power snapshot serves 52,976 features through a tile-based preview while retaining full GeoJSON for provider exports.
- Kept analytical vectors, manual review inputs and WMS reference overlays explicitly separated to prevent unsupported data corrections or provenance loss.

Rachownik | Private Full-Stack Product MVP

Personal project

- Architected a home-budget application with OCR-based receipt tracking using React 19, TypeScript, Vite and a Python/FastAPI backend; deployed the product to a VPS.

Voice-Driven AI RPG | Academic Group Project

Team project

- Collaborated on a voice-controlled game integrating the Gemini API and Google Cloud services for speech input, persistence and deployment; contributed asynchronous data flows and state management.

EDUCATION

Bachelor of Computer Science, in progress | Akademia WSB

2024 - Present

Algorithms, data structures, core computer science fundamentals and cybersecurity.